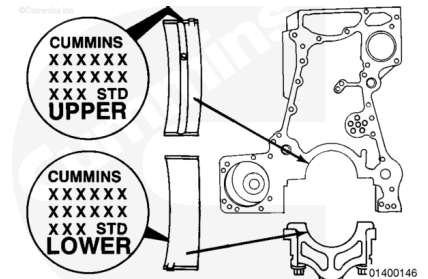


Trainings ▾

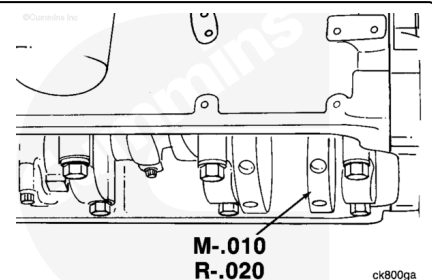
General Information

The upper bearings contain an oil hole. The lower bearings do **not** have an oil hole. Both bearings are marked on the backside indicating location (upper or lower) and size (standard [STD] or oversize [OS]). The amount of the bearing is oversized is indicated in U.S. customary inches.

Use the same size bearing standard, 0.254 mm [0.010 in], 0.508 mm [0.020 in], or 0.762 mm [0.030 in] that was removed.

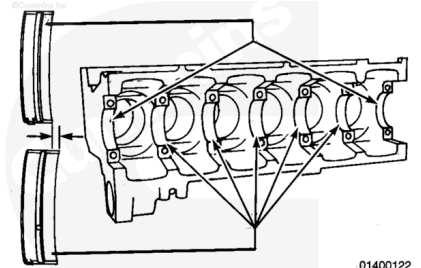


The crankshaft will be stamped on the end of the number one counterweight to indicate if it has been ground undersize. Thrust bearing size is stamped on a crankshaft counterweight, adjacent to the thrust location.



The location number of the main bearing caps, beginning at the front, are numbers 1 through 7.

The main bearings are two widths. The **narrow** main bearings fit locations number 1 and number 7. The **wide** bearings fit the remaining locations.

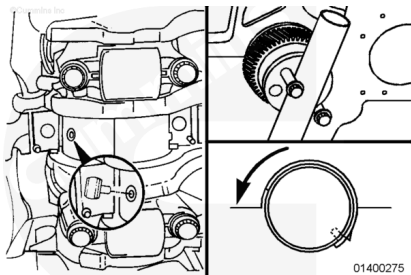


Install

Reused bearings **must** be installed in their original location.

Install the upper main bearing shells using the same method that was used for the removal of the shells.

The bearing tang (1) **must** fit into the slot (2) in the bearing saddle to secure the bearing in the proper location.



Before installing the thrust bearings, Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-006-tr.html>)

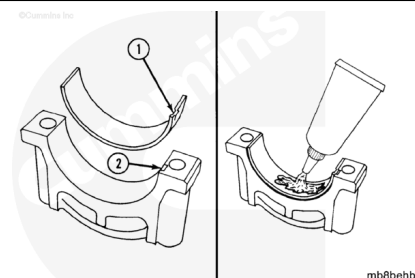
Use a lint-free cloth. Clean the lower main bearings, the lower thrust bearings, and the mounting surfaces.

Do **not** lubricate the back of the main bearings.

Align the tang (1) in the bearing with the slot (2) in the main bearing cap. Install the bearing. The end of the bearing **must** be even with the main bearing cap mounting surface.

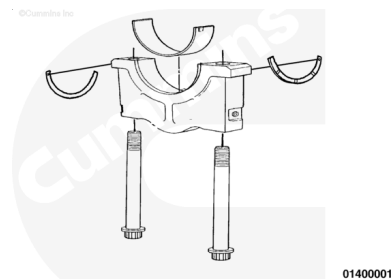
Lubricate the bearing surface with clean engine oil.

Install the upper thrust bearings. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-007-tr.html>)



⚠ CAUTION ⚠

To reduce the possibility of engine damage, the grooves in the thrust bearings must point toward the crankshaft. The dowels that secure the bearings must not protrude above the bearing.



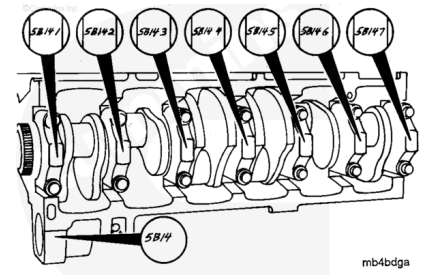
Lubricate the capscrew head and threads with SAE EP 140W oil. Allow the excess oil to drip off of the capscrews before installing in the block.

Install the capscrews in the cap.

Install the two thrust bearings on the number six main bearing cap. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-007-tr.html>)

⚠ CAUTION ⚠

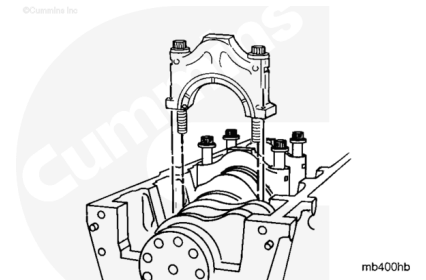
To reduce the possibility of engine damage, the numbers on the main bearing caps must be the same as the numbers on the block.



Check the number on the main bearing caps. The last digit of each number (1 through 7), indicates the correct location.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the side of the cap with the bearing tang slot and the bearing locking tang are toward the same side of the block.

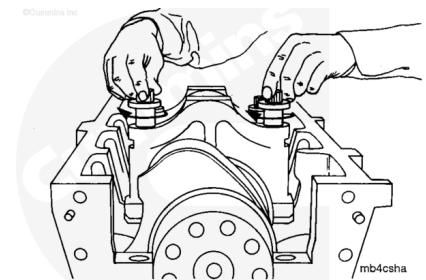


Lubricate the bearing surface with engine oil. The bearing shell **must** be firmly seated in the cap and the correct capscrew, washer combination in position. Install the main bearing caps.

Do **not** hit the main bearing caps with a hammer. The bearing shells can fall out.

⚠ CAUTION ⚠

Do not rotate the crankshaft until all of the main bearing caps are pulled to the block. Damage will result if the bearings move out of location.



Use your hands to install the capscrews.

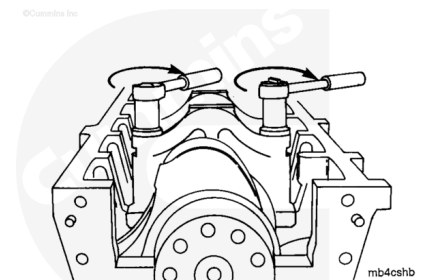
Turn each capscrew until it touches the main bearing caps.

Note : Do not use an impact wrench. The main bearing shells can fall out.

Use both of the capscrews to pull the main bearing cap into position.

Use two wrenches and tighten both capscrews at the same time.

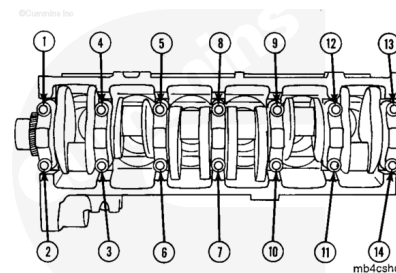
Check to be sure the number 7 cap is even with the back of the block.



Use the following steps and tighten the capscrews in the sequence shown.

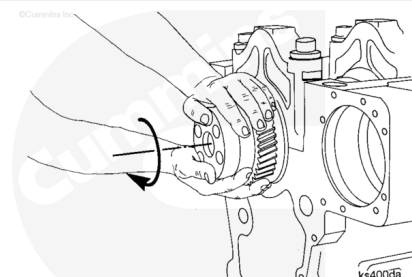
Torque Value:

1. 265 n•m [195 ft-lb]
2. 605 n•m [445 ft-lb]
3. Loosen all.
4. 265 n•m [195 ft-lb]
5. 605 n•m [445 ft-lb]



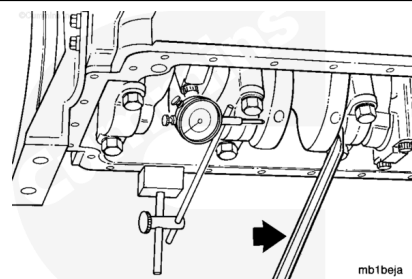
Turn the crankshaft. If the pistons have been removed, use your hands. If the pistons have **not** been removed, use the barring mechanism.

Note : The crankshaft will turn freely if the main bearings are installed correctly.



Use a dial indicator and measure the crankshaft end clearance.

Crankshaft End Clearance		
mm		in
0.10	MIN	0.004
0.41	MAX	0.016



If the clearance is **not** within specifications, check for foreign material. Oversize thrust bearings are available to adjust the end clearance.

Note : The upper and lower thrust bearing on the same side of the main caps must be the same thickness.